Zhiheng Zhang

Trustworthy multi-agent software, operational invariants and trace-based recovery

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Concordia application/student ID: 20814667 | Doctor of Philosophy - Computer Science

International applicant; neither a Canadian citizen nor a permanent resident.

Research profile

I want to study when a coordinator should continue, retry, isolate a component or request confirmation after inconsistent agent outputs. My preparation is in architecture, AI-assisted integration and evidence-backed acceptance of bounded multi-device workflows. I would connect practical trace analysis to a small, explicit set of operational invariants.

Education

University of Melbourne, Australia
Master of Computer Science, 2025-present. Degree and final results pending. Master's research concerns Android privacy evidence; research supervisor: Professor Richard Sinnott.

University of British Columbia, Okanagan campus, Canada
Bachelor of Science in Computer Science, awarded 2024. Undergraduate study completed in English; capstone software engineering project: 93/100.

Selected research and systems experience

Utopia / Digital-City - multi-device systems and acceptance

- Led system design, AI-assisted integration and evidence-backed acceptance of Utopia, a completed and frozen integration of my DS-Hns and Codex Boss projects.
- Bounded physical evidence covers two Windows execution workers, Android/Web control, explicit target-device routing and cross-host result/replay records. Android is a control surface, not a validated third worker.
- Maintained task/evidence links and failure histories for reproducibility. The recorded REX-990 acceptance includes 205 reconstructed runs and 243 resolved trace pointers; these are scoped historical results, not a new test run or proof of general autonomy.
- Implementation stack: JavaScript ES modules/Node.js and Kotlin/Jetpack Compose. Digital-City holds architecture, task and acceptance records; it is not a separate deployed product.

Privacy Lens - Android privacy-evidence research

- Worked on an offline, evidence-first Android research prototype and a curated reproducibility artifact. The project separates admitted observations, technical signals and obligations requiring human review.
- Relevant practice includes versioned evidence admission, reproducible metric checks and claim-to-evidence traceability. The artifact includes TypeScript/React Native/Expo and a Kotlin native layer.
- The work does not establish legal compliance, clinical outcomes, independent detection accuracy or production readiness.

Target: Jamal Bentahar | Concordia Computer Science PhD

Earlier completed projects

- DS-Hns (completed/frozen): durable task state, restart/resume, recovery history and bounded code-task execution.
- Codex Boss (completed/frozen): multi-agent orchestration, inspectable completion evidence and acceptance governance. Both are predecessors of Utopia, not additional independent deployment claims.

Exploratory work and proposed research

- LucidGloves-derived ESP32 work: exploratory firmware and device-integration investigation, supporting an interest in embodied interaction. No validated custom glove or completed user study is claimed.
- Celestial Smart Glass: a documented research proposal for adaptive captions, session-level speaker association and user-controlled cues. No running prototype, measured hardware compatibility, participants or ethics approval is claimed.

Methods and technical practice

- Systems architecture, interface contracts and multi-device integration; designing bounded task lifecycles, state checks and recovery experiments.
- AI-assisted implementation with explicit review and acceptance; Git-based evidence/version management; test design, controlled fault injection, trace analysis and replay.
- Research emphasis: baselines and ablations, independently observable task outcomes, reproducibility and careful separation of simulation, physical observations and proposed work.

Targeted PhD direction

Can a coordinator combining component reliability evidence with current task state reduce cascading failures more effectively than retry rules or trust scores alone, while avoiding excessive isolation of components that are merely slow?

Proposed starting point: Instrument benign multi-step workflows with state/version identifiers, authorization scope and pre/postcondition evidence.

Selected relevant coursework

Software Engineering; Introduction to Discrete Structures; Introduction to Artificial Intelligence; Computer Ethics.

Inspectable project evidence

Utopia - frozen implementation

REX-990 - bounded acceptance record

Privacy Lens - research artifact

Celestial Smart Glass - proposal only

Academic records accompany the email. Melbourne results are current and incomplete, not a final master's transcript. References can be arranged on request.